Spam Filter Performance Metrics Cheat Sheet

Accuracy:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Overall correctness: how many predictions were correct out of all predictions.

Precision (Positive Predictive Value):

Precision = TP / (TP + FP)

Out of all predicted spam emails, how many were actually spam.

Recall (Sensitivity, True Positive Rate):

Recall = TP / (TP + FN)

Out of all actual spam emails, how many were correctly identified as spam.

Specificity (True Negative Rate):

Specificity = TN / (TN + FP)

Out of all actual non-spam emails, how many were correctly identified as non-spam.

F1 Score:

*F1 = 2 * (Precision * Recall) / (Precision + Recall)*

Harmonic mean of precision and recall balances both metrics.

False Positive Rate (Fall-out):

FPR = FP / (FP + TN)

Out of all actual non-spam emails, how many were incorrectly marked as spam.

1 - FPR (True Negative Rate / Specificity):

1 - FPR = TN / (TN + FP)

Also known as specificity; measures how well non-spam is recognized.

False Negative Rate (Miss Rate):

FNR = FN / (FN + TP)

Out of all actual spam emails, how many were missed (marked as non-spam).

ROC Curve:

Plot of TPR vs FPR at various threshold settings

Visual tool to assess trade-off between true positive and false positive rates.

AUC (Area Under Curve):

Area under the ROC curve (between 0 and 1)

Overall ability of the classifier to distinguish between classes.

NPV (Negative Predictive Value):

NPV = TN / (TN + FN)

Out of all predicted non-spam, how many were actually non-spam.

F2 Score:

*F2 = (1 + 2^2) * (Precision * Recall) / (4 * Precision + Recall)*

Gives more weight to recall useful when missing spam is costly.

F0.5 Score:

*F0.5 = (1 + 0.5^2) * (Precision * Recall) / (0.25 * Precision + Recall)*

Gives more weight to precision useful when false alarms are costly.

MCC (Matthews Correlation Coefficient):

MCC = (TPTN FPFN) / sqrt((TP+FP)(TP+FN)(TN+FP)(TN+FN))

Balanced score that works well even with imbalanced data.

Balanced Accuracy:

Balanced Accuracy = (Sensitivity + Specificity) / 2

Average of sensitivity and specificity useful in imbalanced data.

FDR (False Discovery Rate):

FDR = FP / (TP + FP)

Proportion of false spam predictions out of all predicted spam.

FOR (False Omission Rate):

FOR = FN / (TN + FN)

Proportion of missed spam out of all predicted non-spam.

Youdens J Statistic:

J = Sensitivity + Specificity 1

Summarizes performance useful for finding optimal threshold.

Log Loss (Cross-Entropy Loss):

$LogLoss = - (1/N) \; [y \log(p) + (1y) \log(1p)]$

Penalizes incorrect probabilistic predictions more harshly.

Brier Score:

$Brier = (1/N) \; (p \; y)^2$

Measures accuracy of probabilistic predictions lower is better.